Explaining Legal Equation Transformations

Naming the property behind each step, judging equivalence, finding the illegal step, and the inequality sign rule

Directions:

- A transformation is **legal** when the new equation has exactly the same solutions as the old one. Every line you write must be justified by one of the properties below.
- Fill in the **Reason** column of each table as you go, then finish the problem in the work space and write the final answer on the line.
- Where a problem shows someone else's work, name the first line that is *not* justified before you fix anything.

Reason bank (use these names, no values shown).

- **Addition / Subtraction Property of Equality** — add or subtract the same quantity on *both* sides.
- **Multiplication / Division Property of Equality** — multiply or divide *both* sides by the same nonzero quantity.
- **Distributive Property** — $a(b + c) = ab + ac$; rewrites one side without changing its value.
- **Combining like terms** — rewrites one side without changing its value.
- **Not a legal step** — the new equation does not have the same solutions.

1. Solve, and name the reason for each line.

Step	Reason
$3x + 7 = 22$	Given
$3x = 15$	
$x =$ ______	

Now check your solution by substituting it back into $3x + 7$. What do you get?

Answer: ________________

2. Solve, and name the reason for each line. Be careful about which operation you undo first.

Step	Reason
$\frac{x}{4} - 3 = 5$	Given
$\frac{x}{4} = 8$	
$x =$ ______	

Answer: ____________________

3. Rewriting one side is legal only when the two forms are equal for *every* value of the variable. Rewrite $2(x+5)$ without parentheses, name the property that allows it, and explain in one sentence why this rewriting never changes an equation's solutions.

Original side	Rewritten side
$2(x + 5)$	

Answer: ____________________

4. Devi's work is below. Find the first line that is not legal, say what she did instead of a legal step, then solve the equation correctly and check your answer in $5x - 3$.

Devi's step	**Legal? Reason**
$5x - 3 = 12$	Given
$5x = 9$	
$x = \frac{9}{5}$	

Answer: __________________

5. Solve the inequality $-2x < 8$. Write the reason for each line, and state clearly what happens to the inequality symbol and why.

Step	**Reason**
$-2x < 8$	Given
x ____ ______	

Answer: __________________

6. Solve $4(x-2) = 3x + 5$. Write one line per legal step and name the property beside it; the first two rows are started for you.

Step	**Reason**
$4(x-2) = 3x + 5$	Given
$4x - 8 = 3x + 5$	

Answer: ________________

7. A student starts from $x^2 = 4x$ and divides both sides by x to get $x = 4$. Dividing both sides by the same quantity is usually legal — so explain exactly which condition fails here, then find *all* the solutions by moving everything to one side and factoring.

Student's step	**Legal? Reason**
$x^2 = 4x$	Given
$x = 4$	

Answer: ________________

8. Ravi turns $2x + 6 = 10$ into $x + 6 = 5$, saying he "divided by 2". Solve Ravi's new equation, then substitute that value into the *original* left side $2x+6$. Write what the original left side comes out to, and say whether the two equations are equivalent.

Equation	**Its solution**
$x + 6 = 5$	
value of $2x + 6$ there	

Answer: ____________

9. Solve $-3(x - 2) \geq 9$. Show every step with its reason, and mark the exact line where the inequality symbol turns around.

Step	**Reason**
$-3(x - 2) \geq 9$	Given

Answer: ____________

10. Below is a solution with one illegal line. Name the line, name the property it should have used, and then solve $2(x+3) = 4x - 8$ correctly.

Step	**Legal? Reason**
$2(x+3) = 4x - 8$	Given
$2x + 3 = 4x - 8$	
$-2x = -11$	

Answer: __________________